

PRASAD'S PHYSICS

**BRAHMAPUR
Assignment**

Projectile

Numerical Problems

01. What will be the effect on maximum height of a projectile when its angle of projection is changed from 30° to 60° , keeping the same initial velocity of projection?
02. A ball of mass m is thrown vertically up. Another ball of mass $2m$ is thrown at an angle θ with the vertical. Both of them stay in air for the same period of time. What is the ratio of the height attained by the two balls.
03. A ball is thrown horizontally from the top of a tower with a speed of 50 ms^{-1} . Find the velocity and position at the end of 3 second. ($g = 9.8 \text{ ms}^{-2}$)
04. Prove that a gun will shoot three times as high when its angle of elevation is 60° as when it is 30° , but cover the same horizontal range.
05. Prove that the maximum horizontal range is 4 times the maximum height attained by a projectile which is fired along the required oblique direction.
06. A body is projected such that its kinetic energy at the top is $\frac{3}{4}$ th of its initial kinetic energy. What is the initial angle of projection of the projectile with the horizontal?
07. A ball is projected from ground with a velocity of 40 ms^{-1} at an angle of 30° with the horizontal. Determine
 - (a) the time of flight
 - (b) the horizontal range and
 - (c) the maximum height
 - (d) at what time is the maximum height attained?
 - (e) what is the magnitude and direction of velocity at the maximum height?
 - (f) what is the magnitude and direction of velocity after 1s of throwing the ball?
08. A ball is thrown with a velocity of 10 m/s at an angle 45° with horizontal. Find its range.
09. An aircraft is flying horizontally with a velocity of 540 kmh^{-1} at a height of $2,000 \text{ m}$ from ground. When it is vertically above a point A on the ground, a body is dropped from it. The body reaches the ground at a point B. Calculate distance AB. Take $g = 10 \text{ ms}^{-2}$.
10. A ball is thrown vertically upward with velocity 20 m/s from a rail road car moving with a velocity 5 m/s horizontally. A person standing on the ground observes its motion as projectile. Find time of flight and range of the projectile ($g = 10 \text{ m/s}^2$)
11. A particle is projected with $u = 40 \text{ m/s}$ at an angle $\theta = 30^\circ$ with the horizontal from a point on the ground. Find
 - (a) the velocity of the particle after 1 s.
 - (b) the velocity of the particle at the highest point.
 - (c) the maximum height attained.
 - (d) the total time of flight
 - (e) the range of projectile
 - (f) the change in velocities between point of projection and point of impact (if these points are on the ground). (take $g = 10 \text{ m/s}^2$)
12. A helicopter while flying at a height of 100 m with velocity 30 m/s at an angle 30° with the horizontal, drops a packet. Where will the packet strike the ground. ($g = 10 \text{ m/s}^2$)
13. A boy throws a ball horizontally, with a velocity of 8 ms^{-1} , from the top of a building. The ball falls to the ground 12 m away from the building. What is the height of the building?
14. Two tall buildings are 20 m apart. With what speed must a ball be thrown horizontally from a window 100 m above the ground in one building so that it will enter a window 21.6 m above the ground in the other?
15. A body is projected at an angle of 45° to the horizontal. Find the ratio of K.E. at the highest point of the trajectory to the K.E. at the lowest point.
16. A ball is thrown with a velocity of $10\sqrt{2} \text{ ms}^{-1}$ at an angle of projection of 45° to the horizontal. What are velocity and acceleration at the top of trajectory?
17. A boy can throw a ball to a maximum horizontal distance of R . What would be the maximum vertical distance upto which he can throw the ball, keeping the initial velocity same?
18. A ball is projected with kinetic energy K at an angle of 45° to the horizontal. Find the kinetic energy at highest point during its flight.

19. A body is projected at 30° angle with the horizontal with velocity 30 ms^{-1} . What is the angle with the horizontal after 1.5 s ? (Take $g = 10 \text{ ms}^{-2}$).
20. The maximum range of a gun on horizontal terrain is 16 km . If $g = 10 \text{ m/sec}^2$, then muzzle velocity of the shell must be

Multiple Choice Questions

01. If air resistance is not considered in a projectile motion, the horizontal motion takes place with
(a) constant velocity
(b) constant acceleration
(c) constant retardation
(d) variable velocity
02. The point from where a ball is projected is taken as the origin of the coordinate axes. The x and y components of its displacement are given by $x = 6t$ and $y = 8t - 5t^2$. What is the velocity of projection?
(a) 6 m/s (b) 8 m/s
(c) 10 m/s (d) 14 m/s
03. In the above problem, what is the angle of projection with horizontal?
(a) $\tan^{-1}(1/4)$ (b) $\tan^{-1}(4/3)$
(c) $\tan^{-1}(3/4)$ (d) $\tan^{-1}(1/2)$
04. A ball is thrown upwards at an angle of 60° to the horizontal. It falls on the ground at a distance of 90 m . If the ball is thrown with the same initial velocity at an angle 30° , it will fall on the ground at a distance of
(a) 120 m (b) 90 m (c) 60 m (d) 30 m
05. Range of a projectile is R , when the angle of projection is 30° . Then, the value of the other angle of projection for the same range is
(a) 45° (b) 60° (c) 50° (d) 55°
06. A projectile will cover same horizontal distance when the initial angles of projection are
(a) $20^\circ, 60^\circ$ (b) $20^\circ, 50^\circ$
(c) $20^\circ, 40^\circ$ (d) $20^\circ, 70^\circ$
07. Javelin is thrown at an angle of θ with the horizontal and the range is maximum. The value of $\tan \theta$ is
(a) 1 (b) $\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) 2
08. A shot is fired from a point at a distance of 200 m from the foot of a tower 100 m high so that it just passes over it horizontally. The direction of shot with horizontal is
(a) 30° (b) 45° (c) 60° (d) 70°
09. Two bullets are fired horizontally with different velocities from the same height. Which will reach the ground first?

- (a) slower one
(b) faster one
(c) both will reach simultaneously
(d) it cannot be predicted
10. A person can throw a stone to a maximum distance of h meter. The maximum distance to which he can throw the stone is
(a) h (b) $h/2$ (c) $2h$ (d) $3h$
11. The path of one projectile as seen by an observer on another projectile is a/an
(a) straight line (b) parabola
(c) ellipse (d) circle
12. Galileo writes that for angles of projection of a projectile at angles $(45 + \alpha)$ and $(45 - \alpha)$, the horizontal ranges described by the projectile are in the ratio of
(a) $2 : 1$ (b) $1 : 2$
(c) $1 : 1$ (d) $2 : 3$
13. At the top of the trajectory of a projectile, the acceleration is
(a) maximum (b) minimum
(c) zero (d) g
14. At the uppermost point of a projectile, its velocity and acceleration are at an angle of
(a) 0° (b) 45° (c) 90° (d) 180°
15. An airplane moving horizontally with a speed of 180 km/h drops a food packet while flying at a height of 490 m . The horizontal range of the food packet is ($g = 9.8 \text{ m/s}^2$)
(a) 180 m (b) 980 m
(c) 500 m (d) 670 m
16. A gun fires two bullets at 60° and 30° with the horizontal. The bullets strike at some horizontal distance. The ratio of the maximum height for the two bullets is in the ratio
(a) $2 : 1$ (b) $3 : 1$
(c) $4 : 1$ (d) $1 : 1$
17. When a projectile is thrown horizontally, the resultant velocity of the projectile at any time t is given by
(a) gt (b) $\frac{1}{2}gt^2$
(c) $\sqrt{u^2 + g^2t^2}$ (d) $\sqrt{u^2 - g^2t^2}$
18. A projectile is projected with initial velocity $(6\hat{i} + 8\hat{j}) \text{ m/s}$. If $g = 10 \text{ ms}^{-2}$, then horizontal range is
(a) 4.8 m (b) 9.6 m (c) 19.2 m (d) 14.0 m
19. A ball is thrown at different angles with the same speed u and from the same point and it has the same range in both the cases. If y_1 and y_2 be the

heights attained in the two cases, then $y_1 + y_2$ is equal to

- (a) $\frac{u^2}{g}$ (b) $\frac{2u^2}{g}$ (c) $\frac{u^2}{2g}$ (d) $\frac{u^2}{4g}$

20. A bullet is fired horizontally from a rifle at a distance target. Ignoring the effect of air resistance, which of the following is correct ?

Horizontal Acceleration	Vertical Acceleration
(a) 10 ms^{-2}	10 ms^{-2}
(b) 10 ms^{-2}	0 ms^{-2}
(c) 0 ms^{-2}	10 ms^{-2}
(d) 0 ms^{-2}	0 ms^{-2}